

UBC ELEC 221: Introduction to Signals and Systems

Continuous Time Signals

Even and Odd Components

$$x_{\text{even}}(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_{\text{odd}}(t) = \frac{1}{2}[x(t) - x(-t)]$$

Periodicity

$$x(t + kT_0) = x(t) \forall t \in (-\infty, \infty)$$

Energy

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Power

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

Causality:

- Causal: $x(t) = 0$ for $t < 0$.
- Anti-Causal: $x(t) = 0$ for $t \geq 0$.
- Acausal or Non-Causal: Both of the above.

Continuous Time Systems

Linearity

$$\mathcal{S}[\alpha x(t) + \beta y(t)] = \alpha \mathcal{S}[x(t)] + \beta \mathcal{S}[y(t)]$$

Time Invariance

$$\mathcal{S}[x(t)] = y(t) \Rightarrow \mathcal{S}[x(t \pm \tau)] = y(t \pm \tau)$$

Zero-State Response. Due to the input as the initial conditions are zero.

Zero-Input Response. Due to the initial conditions as the input is zero.

Convolution

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau = \int_{-\infty}^{\infty} x(t - \tau)h(\tau) d\tau$$

Causality. A continuous time system $\mathcal{S}$ is causal if whenever $x(t) = 0$ and there are no initial conditions, $y(t) = 0$ and the output $y(t)$ does not depend on future inputs.

BIBO Stability. If an input bounded then the output of a BIBO stable system is bounded.

$$\Leftrightarrow \int_{-\infty}^{\infty} |h(t)| dt < \infty$$

Laplace Transform

Eigenfunction Property

$$\mathcal{S}[e^{s_0 t}] = H(s_0)e^{s_0 t}$$

One Sided Laplace Transform:

$$F(s) = \mathcal{L}[f(t)u(t)] = \int_{0^-}^{\infty} f(t)e^{-st} dt, \forall s = \sigma + j\omega$$

Note: $|H(s)|$ is greatest when s is closest to the poles.

Signal Support ROC

Finite support	Entire s-plane.
Causal function	$\sigma > \max(\sigma_i), -\infty < \omega < \infty$
Anti-causal	$\sigma < \min(\sigma_i), -\infty < \omega < \infty$
Non-causal	$\mathcal{R} = \mathcal{R}_{\text{causal}} \cap \mathcal{R}_{\text{anti-causal}}$

Initial Value Theorem

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s)$$

Final Value Theorem

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

BIBO Stability. If the region of convergence contains the $j\omega$ -axis then the system is BIBO stable.

2-Sided LT from 1-Sided LT

- For an anti-causal function $C(s) = \mathcal{L}[c(-t)u(t)]_{(-s)}$
- For a 2-sided function
 $p(t) = p_{ac}(t) + p_c(t) = p(t)u(-t) + p(t)u(t)$
- $P(t) = \mathcal{L}[p_{ac}(-t)u(t)]_{(-s)} + \mathcal{L}[p_c(t)u(t)]$

CT Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t}$$

Eigenfunction Property

$$\mathcal{S}[e^{j\omega_0 t}] = H(j\omega_0)e^{j\omega_0 t}$$

$$x(t) = \sum_k X_k e^{j\omega_k t}$$

$$\begin{aligned} \Rightarrow y(t) &= \sum_k X_k H(j\omega_k) e^{j\omega_k t} \\ &= \sum_k X_k |H(j\omega_k)| e^{j(\omega_k t + \angle H(\omega_k))} \end{aligned}$$

Fourier Series Coefficients (for any t_0)

$$X_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\omega_0 t} dt$$

Parseval's Power Relation (for any t_0):

$$P = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |X_k|^2$$

Symmetry of Line Spectra:

$$\begin{aligned} |X_k| &= |X_{-k}| \\ \angle X_k &= -\angle X_{-k} \end{aligned}$$

Trigonometric Fourier Series:

$$x(t) = X_0 + 2 \sum_{k=1}^{\infty} |X_k| \cos(k\omega t + \Theta_k)$$

$$x(t) = c_0 + 2 \sum_{k=1}^{\infty} [c_k \cos(k\omega_0 t) + d_k \sin(k\omega_0 t)]$$

$$c_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) \cos(k\omega_0 t) dt \quad k = 0, 1, 2, \dots$$

$$d_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) \sin(k\omega_0 t) dt \quad k = 1, 2, 3, \dots$$

$$\Theta_k = -\arctan(d_k/c_k)$$

Fourier Coefficients from Laplace Transform. If $x_1(t)$ is a single period of $x(t)$, then

$$X_k = \frac{1}{T_0} \mathcal{L}[x_1(t)] \big|_{s=jk\omega_0}$$

Response of LTI Systems to Periodic Signals. If the input to an LTI system has Fourier Series $x(t) = X_0 + 2 \sum_{k=1}^{\infty} |X_k| \cos(k\omega t + \angle X_k)$, then the steady state response is

$$y(t) = X_0 |H(j\omega)| + 2 \sum_{k=1}^{\infty} |X_k| |H(jk\omega_0)| \cos(k\omega_0 t + \angle X_k + \angle H(jk\omega_0))$$

CT Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Fourier Transform from Laplace Transform (if $X(s)$ contains the $j\omega$ -axis),

$$\mathcal{F}[x(t)] = \mathcal{L}[x(t)] \big|_{s=j\omega} = X(s) \big|_{s=j\omega}$$

Fourier Transform of Periodic Signals under Fourier Series

$$\sum_k X_k e^{jk\omega_0 t} \Leftrightarrow \sum_k 2\pi X_k \delta(\omega - k\omega_0)$$

Parseval's Energy Relation

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Basic Properties of Laplace Transforms

Property	t-domain	s-domain
Linearity	$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
1st Derivative	$f'(t)$	$sF(s) - f(0^-)$
2nd Derivative	$f''(t)$	$s^2 F(s) - sf(0^-) - f'(0^-)$
Frequency Differentiation	$(-t)^n f(t)$	$\frac{d^n}{ds^n} F(s)$
Integration	$\int_{0^-}^t f(\tau) d\tau$	$\frac{F(s)}{s}$
Frequency Integration	$\frac{f(t)}{t}$	$\int_{-\infty}^{-s} F(-\tau) d\tau$
Convolution	$\int_0^t f(t-\tau)g(\tau) d\tau$	$G(s)F(s)$
Time-shifting	$f(t-\tau)$	$e^{-\tau s} F(s)$
Frequency-shift	$e^{at} f(t)$	$F(s-a)$
Time Scaling	$f(at)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$

One-sided Laplace Transforms

f(t)	F(s)	Region of Convergence
$\delta(t)$	e^{-as}	whole s-plane
$u(t)$	$\frac{1}{s}$	$\text{Re}[s] > 0$
$tu(t)$	$\frac{1}{s^2}$	$\text{Re}[s] > 0$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\text{Re}[s] > 0$
$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$	$\text{Re}[s] > 0$
$\cos(\omega t)u(t)$	$\frac{s}{s^2 + \omega^2}$	$\text{Re}[s] > 0$
$t \sin(\omega_0 t)u(t)$	$\frac{2\omega_0 s}{(s^2 + \omega_0^2)^2}$	$\text{Re}[s] > 0$
$t \cos(\omega_0 t)u(t)$	$\frac{s^2 - \omega_0^2}{(s^2 + \omega_0^2)^2}$	$\text{Re}[s] > 0$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\text{Re}[s] > -a$
te^{-at}	$\frac{1}{(s+a)^2}$	$\text{Re}[s] > -a$
$t^n e^{-at}u(t)$	$\frac{n!}{(s+a)^{n+1}}$	$\text{Re}[s] > -a$
$e^{-\alpha t} \sin(\beta t)u(t)$	$\frac{\beta}{(s+\alpha)^2 + \beta^2}$	$\text{Re}[s] > -\alpha$
$e^{-\alpha t} \cos(\beta t)u(t)$	$\frac{s+\alpha}{(s+\alpha)^2 + \beta^2}$	$\text{Re}[s] > -\alpha$

Two-sided Laplace Transforms

f(t)	F(s)	Region of Convergence
$u(-t)$	$-\frac{1}{s}$	$\text{Re}[s] < 0$
$e^{at}u(-t)$	$-\frac{1}{s-a}$	$\text{Re}[s] < a$
$e^{at}u(t) + e^{bt}u(-t)$	$\frac{1}{s-a} - \frac{1}{s-b}$	$b < \text{Re}[s] < a$
$e^{-a t }, a > 0$	$\frac{2a}{s^2 - a^2}$	$-a < \text{Re}\{s\} < a$
$\text{sgn}(t)$	$\frac{2}{s}$	$\text{Re}[s] \neq 0$

Trigonometric Value

Angle (°)	Angle (rad)	sin θ	cos θ	tan θ
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	undefined

Basic Properties of Fourier Series

Property	t-domain	ω -domain
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X_k + \beta Y_k$
Power	$\frac{1}{T} \int_0^T x(t) ^2 dt$	$\sum_{n=-\infty}^{\infty} X_k ^2$
Differentiation	$x'(t)$	$jk\omega_0 X_k$
Integration (only if $X_0 = 0$)	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{X_k}{jk\omega_0}, k \neq 0$
Time-shifting	$x(t-\alpha)$	$e^{-j\alpha\omega_0 k} X_k$
Frequency-shift	$e^{j\alpha\omega_0 t} x(t)$	$X_{k-\alpha}, \alpha \in \mathbb{Z}$
Convolution	$z(t) = [x * y](t)$	$X_k Y_k$
Symmetry	x(t) real	$ X_k = X_{-k} $ even $\angle X_k = -\angle X_{-k}$ odd

Basic Properties of CT Fourier Transforms

Property	t-domain	ω -domain
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X(\omega) + \beta Y(\omega)$
Time Scaling	$x(\alpha t), \alpha \neq 0$	$\frac{1}{ \alpha } X\left(\frac{\omega}{\alpha}\right)$
Energy	$\int_{-\infty}^{\infty} x(t) ^2 dt$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega$
Reflection	$x(-t)$	$X(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Differentiation	$\frac{d^n}{dt^n} x(t)$	$(j\omega)^n X(\omega)$
Frequency Differentiation	$(-jt)^n x(t)$	$\frac{d^n}{d\omega^n} X(\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Time-shifting	$x(t-\alpha)$	$e^{-j\alpha\omega} X(\omega)$
Frequency-shift	$e^{j\omega_0 t} x(t)$	$X(\omega - \omega_0)$
Modulation	$x(t) \cos(\omega_0 t)$	$\frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$
Periodic signals	$\sum X_k e^{j\omega_0 t}$	$\sum 2\pi X_k \delta(\omega - k\omega_0)$
Symmetry	x(t) real	$ X(\omega) = X(-\omega) $ even $\angle X(\omega) = -\angle X(-\omega)$ odd
Windowing	$x(t)y(t)$	$\frac{1}{2\pi} [X * Y](\omega)$
Cosine Transform	$x(t)$ even	$\int_{-\infty}^{\infty} x(t) \cos(\omega t) dt$
Sine Transform	$x(t)$ odd	$-j \int_{-\infty}^{\infty} x(t) \sin(\omega t) dt$
Trigonometric Formulae		
Angle Sum/Difference	$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$	
	$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$	
Double Angle	$\tan(a \pm b) = \frac{\tan a \pm \tan b}{1 \mp \tan a \tan b}$	
	$\sin(2\theta) = 2 \sin \theta \cos \theta$	
	$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$	
	$\cos(2\theta) = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$	
Product-to-Sum	$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$	
	$\sin a \sin b = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$	
	$\cos a \cos b = \frac{1}{2} [\cos(a-b) + \cos(a+b)]$	
	$\sin a \cos b = \frac{1}{2} [\sin(a+b) + \sin(a-b)]$	

CT Fourier Transform Pair

$\mathbf{x(t)}$	$\mathbf{X(\omega)}$	
1	$2\pi\delta(\omega)$	
$\delta(t)$	1	
$\delta(t-\tau)$	$e^{-j\omega\tau}$	
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$	
$u(-t)$	$\frac{-1}{j\omega} + \pi\delta(\omega)$	
$\text{sgn}(t) = 2[u(t) - 0.5]$	$\frac{2}{j\omega}$	
$e^{-at}u(t)$	$\frac{1}{a + j\omega}$	$a > 0$
$e^{at}u(-t)$	$\frac{1}{a - j\omega}$	$a > 0$
$te^{at}u(t)$	$\frac{1}{(a + j\omega)^2}$	$a > 0$
$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$	$a > 0$
$\cos(\omega_0t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
$\sin(\omega_0t)$	$-j\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	
$u(t + \tau) - u(t - \tau)$	$2\tau \frac{\sin(\omega\tau)}{\omega\tau},$	$\tau > 0$
$\frac{\sin(\omega_0t)}{\pi t}$	$u(\omega + \omega_0) - u(\omega - \omega_0)$	
$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$	$\omega_0 = \frac{2\pi}{T}$

Sampling Theory

$$x_s(t) = x(nT_s) = x(t)|_{t=nT_s} = x(t) \sum_n \delta(t - nT_s)$$

$$X_s(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s)$$

Nyquist-Shannon Sampling Rate

$$\omega_s = \frac{2\pi}{T_s} \geq 2\omega_{\max}$$

Aliasing occurs if $\omega_s < 2\omega_{\max}$

Aliasing frequency mapping for $[0, f_s/2]$

$$f_{\text{alias}} = |f - kf_s|$$

Reconstruction $X(\omega) = X_s(\omega)H_{\text{lp}}(\omega)$:

$$H_{\text{lp}}(\omega) = \begin{cases} T_s & -\frac{\omega_s}{2} \leq \omega \leq \frac{\omega_s}{2} \\ 0 & \text{otherwise} \end{cases}$$

Inclusive bounds. If represented by a unit step function the bounds are inclusive for filters

Signal Reconstruction from Sinc Interpolation

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \frac{\sin(\pi(t - nT_s)/T_s)}{\pi(t - nT_s)/T_s}$$

Discrete Time Signals

Definition $x[n] = x(nT_0)$.

Periodicity

$$x[n + kN] = x[n] \quad \forall k \in \mathbb{Z}$$

When sampling an analog sinusoid of fundamental period T_0 , we obtain a periodic discrete sinusoid provided that m, N not divisible by each-other,

$$\frac{T_s}{T_0} = \frac{m}{N}$$

Aliasing occurs if

$$T_s > T_0/2$$

Energy

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Power

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N |x[n]|^2$$

Convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

Bounded-Input Bounded-Output Stability

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Discrete Time Systems

Linearity

$$\mathcal{S}[\alpha x[n] + \beta y[n]] = \alpha \mathcal{S}[x[n]] + \beta \mathcal{S}[y[n]]$$

Time Invariance

$$\mathcal{S}[x[n]] = y[n] \Rightarrow \mathcal{S}[x[n \pm \tau]] = y[n \pm \tau]$$

Convolution

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

BIBO Stability. If an input bounded then the output of a BIBO stable system is bounded.

$$\Leftrightarrow \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Z-Transform

$$\mathcal{Z}[x(nT_s)] = \mathcal{L}[x_s(t)]|_{z=e^{sT_s}} = \sum_n x(nT_s)z^{-n}$$

Convergence

$$|X(z)| = \sum_n |x[n]|r^{-n} < \infty$$

Initial Value Theorem

$$x[0] = \lim_{z \rightarrow \infty} X(z)$$

Final Value Theorem

$$\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (z - 1)X(z)$$

BIBO Stability. If the ROC contains radius $z = 1$, then the system is BIBO stable.

DT Fourier Series

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{jk\Omega_0 n}$$

Fourier Series Coefficients (for any t_0)

$$X_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n]e^{-jk\Omega_0 n} \quad \text{where } \Omega_0 = 2\pi/N$$

Basic Properties of Z-Transforms

Property	t-domain	z-domain
Linearity	$\alpha x[n] + \beta y[t]$	$\alpha X(z) + \beta Y(z)$
Convolution	$(x * y)[n]$ $= \sum_k x[k]y[n - k]$	$X(z)Y(z)$
Time-delay (Causal)	$x[n - N]$ $(N \in \mathbb{Z})$	$z^{-N}X(z)$
Time-delay (Causal)	$x[n - N]$ $(N \in \mathbb{Z})$	$z^{-N}X(z) + \sum_{k=1}^N x[-k]z^{-N+k}$
Time reversal	$x[-n]$	$X(z^{-1})$
Multiplication by n	$nx[n]$	$-z \frac{d}{dz} X(z)$
Multiplication by n^2	$n^2x[n]$	$z^2 \frac{d^2}{dz^2} X(z) + z \frac{d}{dz} X(z)$
Finite Difference	$x[n] - x[n - 1]$	$(1 - z^{-1})X(z) - x[-1]$
Accumulation	$\sum_{k=0}^n x[k]$	$\frac{X(z)}{1 - z^{-1}}$

One-sided Z-Transforms

$\mathbf{x[n]}$	$\mathbf{X(z)}$	Region of Convergence
$\delta[n]$	1	whole z -plane
$\delta[n - n_0]$	z^{-n_0}	$z \neq 0, n_0 \geq 1$
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$nu[n]$	$\frac{z^{-1}}{(1 - z^{-1})^2}$	$ z > 1$
$n^2 u[n]$	$\frac{z^{-1}(1 + z^{-1})}{(1 - z^{-1})^3}$	$ z > 1$
$\alpha^n u[n], \alpha < 1$	$\frac{1}{1 - \alpha z^{-1}}$	$ z > \alpha $
$n\alpha^n u[n], \alpha < 1$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$ z > \alpha $
$\cos(\omega_0 n)u[t]$	$\frac{1 - \cos(\omega_0)z^{-1}}{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}$	$ z > 1$
$\sin(\omega_0 n)u[t]$	$\frac{\sin(\omega_0)z^{-1}}{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}$	$ z > 1$
$\alpha^n \cos(\omega_0 n)u[t]$ $ \alpha < 1$	$\frac{1 - \alpha \cos(\omega_0)z^{-1}}{1 - 2\alpha \cos(\omega_0)z^{-1} + \alpha^2 z^{-2}}$	$ z > \alpha $
$\alpha^n \sin(\omega_0 n)u[t]$ $ \alpha < 1$	$\frac{\alpha \sin(\omega_0)z^{-1}}{1 - 2\alpha \cos(\omega_0)z^{-1} + \alpha^2 z^{-2}}$	$ z > \alpha $

Two-sided Z Transforms

$\mathbf{x[n]}$	$\mathbf{X(z)}$	Region of Convergence
$-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
$-\alpha^n u[-n - 1]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z < \alpha $
$-n\alpha^n u[-n - 1]$	$\frac{1}{1 - \alpha z^{-1}} - \frac{1}{1 - \alpha^{-1}z^{-1}}$	$ \alpha < z < \frac{1}{ \alpha }$

Basic Properties of DT Fourier Transforms

Property	t-domain	Ω-domain
Z-transform	$x[n], X(z),$ $ z = 1 \in \text{ROC}$	$X(z = e^{j\Omega})$
Periodicity	$x[n]$	$X(e^{j(\Omega + 2\pi k)})$
Time-shifting	$x[n - N]$	$e^{-j\Omega N} X(e^{j\Omega})$
Frequency-shift	$x[n]e^{j\Omega_0 n}$	$X(e^{j(\Omega - \Omega_0)})$
Convolution	$(x * y)[n]$	$X(e^{j\Omega})Y(e^{j\Omega})$
Multiplication	$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\Omega - \theta)})d\theta$
Symmetry	$x[n] \in \mathbb{R}$	$ X(e^{j\Omega}) $ even function of Ω $\angle X(e^{j\Omega})$ odd function of Ω
Power	$\sum_{n=-\infty}^{\infty} x[n] ^2$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) ^2 d\Omega$

Discrete-Time Fourier Transform (DTFT)

$\mathbf{x[n]}$	$\mathbf{X(e^{j\Omega})}$	
$\delta[n]$	1	$ \Omega < \pi$
1,	$2\pi\delta(\Omega)$	$ \Omega < \pi$
$e^{j\Omega_0 n}$	$2\pi\delta(\Omega - \Omega_0)$	$ \Omega < \pi$
$\alpha^n u[n], \alpha < 1$	$\frac{1}{(1 - \alpha e^{-j\Omega})^2}$	$ \Omega < \pi$
$\cos(\omega_0 n)u[n]$	$\pi[\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)]$	$ \Omega < \pi$
$\sin(\omega_0 n)u[n]$	$\frac{\pi}{j}[\delta(\Omega - \Omega_0) - \delta(\Omega + \Omega_0)]$	$ \Omega < \pi$
$\alpha^n n , \alpha < 1$	$\frac{1 - \alpha^2}{1 - 2\cos(\Omega) + \alpha^2}$	$ \Omega < \pi$
$u[n + \frac{N}{2}] - u[n + \frac{N}{2}]$	$\frac{\sin(\Omega(N + 1)/2)}{\sin(\Omega/2)}$	$ \Omega < \pi$
$\alpha^n \cos(\omega_0 n)u[n]$	$\frac{1 - \alpha \cos(\Omega_0)e^{-j\Omega}}{1 - 2\alpha \cos(\Omega_0)e^{-j\Omega} + \alpha^2 e^{-2j\Omega}}$	
$\alpha^n \sin(\omega_0 n)u[n]$	$\frac{\alpha \sin(\Omega_0)e^{-j\Omega}}{1 - 2\alpha \cos(\Omega_0)e^{-j\Omega} + \alpha^2 e^{-2j\Omega}}$	

State Space Representation

Continuous Time

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

Transfer function matrix for s -domain

$$\mathbf{H}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

Discrete Time

$$\mathbf{x}[n + 1] = \mathbf{A}\mathbf{x}[n] + \mathbf{B}\mathbf{u}[n]$$

$$\mathbf{y}[n] = \mathbf{C}\mathbf{x}[n] + \mathbf{D}\mathbf{u}[n]$$

Transfer function matrix for z -domain

$$\mathbf{H}(z) = \mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

Diagonalization

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}, \quad \mathbf{D} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix}$$

Similarity Transformation

$$\tilde{\mathbf{x}}(t) = \mathbf{T}\mathbf{x}(t)$$

$$[\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}] \Rightarrow [\tilde{\mathbf{A}} = \mathbf{T}\mathbf{A}\mathbf{T}^{-1}, \tilde{\mathbf{B}} = \mathbf{T}\mathbf{B}, \tilde{\mathbf{C}} = \mathbf{C}\mathbf{T}^{-1}, \tilde{\mathbf{D}} = \mathbf{D}]$$

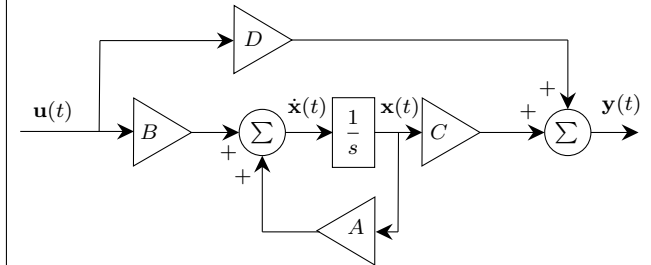
Controllable Canonical Form

$$(\mathbf{A}, \mathbf{B}) \text{ is controllable} \Leftrightarrow \text{rank}[\mathbf{B}, \mathbf{A}\mathbf{B}, \dots, \mathbf{A}^{N-1}\mathbf{B}] = N$$

Observable Canonical Form

$$(\mathbf{A}, \mathbf{C}) \text{ is observable} \Leftrightarrow \text{rank} \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \vdots \\ \mathbf{C}\mathbf{A}^{N-1} \end{bmatrix} = N$$

State-Space Block Diagram



Matrix exponential

$$e^{\mathbf{A}t} = \mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1}, \quad e^{\mathbf{D}t} = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix}$$

Cayley-Hamilton Theorem

$$p(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A}) \Rightarrow p(\mathbf{A}) = 0$$

Miscellaneous Identities

Geometric Series

$$\sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x}, \quad \sum_{k=0}^{\infty} x^k = \frac{1}{1 - x} \quad (0 < x < 1)$$

Euler's Formula

$$e^{j\theta} = \cos \theta + j \sin \theta, \quad \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Problem